

1.1.7 Universal Logic Gates (NAND and NOR)

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR.

CS XI

Chapter 1: Computer Systems

Lecture 08 of 20

1.1.7 Universal Logic Gates (NAND and NOR)

LEARN IT

NAND means "NOT AND": it is an AND gate followed by a NOT. **NOR** means "NOT OR": an OR gate followed by a NOT. On the symbol, the NOT is shown as a small bubble at the output.

NAND and NOR are called **universal gates** because **any logic circuit in the world can be built using only NAND gates (or only NOR gates)**. Factories like this: they can mass-produce one type of chip and build everything from it.



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

NAND (NOT AND) — $Y = (A \cdot B)'$

A	B	$A \cdot B$	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOR (NOT OR) — $Y = (A + B)'$

A	B	$A + B$	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

WORKED EXAMPLE

Show that a NAND gate can work as a NOT gate.

Step 1: Connect the same input A to both inputs of the NAND gate. Then $Y = (A \cdot A)'$.

Step 2: Since $A \cdot A = A$, we get $Y = A'$.

Step 3: Check: $A = 0 \rightarrow (0 \cdot 0)' = 0' = 1$. $A = 1 \rightarrow (1 \cdot 1)' = 1' = 0$. This is exactly the NOT truth table.

In the same way, a NOR gate with both inputs tied together gives $(A + A)' = A'$. And a NAND followed by a NAND-as-NOT gives $((A \cdot B)')' = A \cdot B$, an AND gate. This is why they are "universal".

REMEMBER IT

NAND is the exact **opposite** of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

COMMON MISTAKE

Do not forget the bubble. Without the small circle, a NAND symbol becomes an AND symbol and your answer is wrong. Also, $(A \cdot B)'$ is **not** the same as $A' \cdot B'$: for $A = 1, B = 0$, $(1 \cdot 0)' = 1$ but $0 \cdot 1 = 0$.

اردو میں سمجھیں

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR۔ انہیں یونیورسل (عالمگیر) گیٹ کہتے ہیں کیونکہ صرف NAND یا صرف NOR گیٹس سے دنیا کا کوئی بھی لاجک سرکٹ بنایا جا سکتا ہے۔ NAND صرف تب 0 دیتا ہے جب دونوں ان پٹ 1 ہوں، اور NOR صرف تب 1 دیتا ہے جب دونوں ان پٹ 0 ہوں۔

CHECK YOURSELF

1. Why are NAND and NOR called universal gates?
2. What is the NOR output for $A = 0, B = 1$?
3. How do you make a NOT gate from a NAND gate?

Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.

Next lecture: 1.1.7 Advanced Logic Gates (XOR and XNOR)